

When Camera Confirmation Helps and Hurts: Radar–Camera AEB Permission Policies and Failure Severity in CARLA

Mai Viet Hoang and Duc An Pham
Hanoi University of Science and Technology
Hanoi, Vietnam
hoangmai04222@gmail.com

Abstract—A car-only camera detector can suppress radar-triggered false braking, but failure to recognize a vehicle does not establish free road. We evaluate three late brake-permission policies—radar-only, hard car-camera gating, and car camera gating with radar emergency fallback—in a shared sensor-to-brake CARLA pipeline. The primary unit is a named condition: 105 core conditions and a 14-condition frozen adverse hold-out, each repeated five times to assess consistency, not traffic prevalence. Core precision/recall is 0.913/0.988, 1.000/0.965, and 1.000/0.988, respectively. Fallback recovers two central non-vehicle hazards rejected by hard gating while suppressing eight edge-prop false brakes. The ordering reverses on hold-out: hard gating passes 11/14 conditions versus 7/14 for fallback, because four persistent synthetic-ghost conditions satisfy the emergency rules. All 20 fallback ghost runs stop below 1 km/h from a median onset of 78.4 km/h, with 3.60-s braking and 9.02 m/s² median peak deceleration. A previously unused hold-out bench condition exposes the opposite cost: last-tick pre-impact speed is 32.57 km/h for radar-only, 54.93 km/h for fallback, and 59.95 km/h for hard gating. Paired outcomes, component ablations, controlled perturbations, and traceable logs from a frozen 2,461-run CUDA campaign locate failures at track formation, permission, and post-permission stopping. The contribution is a controlled empirical comparison of brake policies and failure severity, not a new fusion primitive or a general safety advantage.

Index Terms—automatic emergency braking, camera gating, radar emergency fallback, false braking, fault injection, scenario-based evaluation

I. INTRODUCTION

Automatic emergency braking (AEB) must decide not only whether an object was detected, but whether the available evidence justifies a timely brake. Field data associate AEB with fewer front-to-rear crashes [1], and production designs balance intervention against unwanted activation [2]. Radar provides range and relative velocity; camera semantics can reject implausible radar targets. However, a car-only camera detector may also withhold permission for a genuine non-vehicle obstacle. AEB performance consequently depends on the perception–risk–actuation chain rather than detector accuracy alone [3]; such fault-free perception insufficiencies are the concern of SOTIF [4].

This study isolates the *late brake-permission policy*: the rule that accepts or vetoes a brake already requested by a shared radar risk stage. Hard camera gating requires visual confirmation. Emergency fallback relaxes that veto only for

stable, central, well-supported radar tracks at critical TTC and stopping margin. The intended benefit is selective recovery rather than a return to unrestricted radar-only braking. The unresolved question is whether the same rule can distinguish an unlabelled obstacle from a convincing ghost.

We ask: (RQ1) which named conditions change system outcome when the permission policy changes; (RQ2) which failure mechanisms survive an emergency override; and (RQ3) what is hidden when evaluation stops at brake recall and collision counts? CARLA enables controlled substitution and repeatable near-collision experiments [5], [6]. It does not make the constructed scenario mix representative of public-road traffic.

Our contributions are: (1) a paired, three-policy comparison with the radar processing, controller, and scoring held common; (2) a mechanism-oriented analysis linking camera veto, emergency qualification, and missing tracks to core and adverse hold-out outcomes; and (3) scenario-level reporting coupled to false-brake and pre-impact severity, supported by a frozen CUDA execution and artifact trail. These contributions are empirical. The additional organization and paired reporting in this revision reuse the existing frozen evidence; they do not introduce a new algorithm or a new campaign.

II. RELATED WORK AND RESEARCH BOUNDARY

Radar–vision vehicle recognition for AEB is established. Kim and Song address false target selection using shape and motion evidence [7]; Hsu *et al.* filter radar/camera noise including reflection-related returns [8]. Bae *et al.* use LiDAR-camera fusion for closest-in-path vehicle estimation [9]. Zhang *et al.* combine curved-road target selection, TTC, safety distance, and graded braking [10]. Learned radar–camera detectors such as CRF-Net [11], radar–vision distant-vehicle detection [12], and CenterFusion [13] improve object detection; they do not decide brake permission. We therefore claim neither camera confirmation nor graded braking as new.

The closest verification-oriented work is the preprint by Akula *et al.* [14]. It tests radar-guided edge-density verification on an instrumented vehicle rather than full-frame semantic detection. Its staged results report no missed brakes in 33 threat sessions and braking in all three no-threat sessions at the deployed high-recall operating point. This reinforces the

TABLE I
SHARED, FROZEN SENSOR AND DETECTOR CONFIGURATION.

Item	Setting
Camera	RGB, 1280×720, 70° FOV; $(x, z) = (0.43, 1.35)$ m
Radar	100 m, 30° × 6°, 2000 points/s; $(x, z) = (2.53, 0.48)$ m
Detector	YOLO26n ONNX [28], 640 input; confidence 0.25; NMS IoU 0.50
Dataset	1505/300/200 images; 28/6/4 disjoint train/val/test collection sessions
Training	One <i>car</i> class; AdamW; seed 2026; model SHA-256 prefix dc0a6ca1754a
Timing	Both sensors: 0.05 s; inference: 0.15 s; confirmation hold: 0.35 s; CUDA required

need to inspect unwanted interventions, but is not a numerical baseline for our CARLA results: sensing, speeds, labels, and verification methods differ. Our narrower comparison holds the upstream pipeline fixed and substitutes three brake-permission rules, including a radar override of a car-only gate.

Riedmaier *et al.* survey scenario-based safety assessment [15]; simulation-based pipeline evaluation [16] and alternative visual sensing for CARLA AEB [17] are also prior work. Rao *et al.* reconstruct non-standard AEB corner cases and quantitatively analyse failures [18]; corner-case evaluation itself is not our novelty. Kraus *et al.* provide annotated automotive radar ghosts [19]. Unlike measured multipath, our synthetic returns are rule-directed fault injections, as in simulation fault-injection studies of autonomous driving [20]. Radar-model abstraction strongly limits transfer from virtual tests [21].

Probabilistic threat assessment [22] and Bayesian object-existence estimation in radar-camera tracking [23] further bound the contribution. Our fallback is a deterministic permission rule, not calibrated existence inference. The research question is what this simple rule gains, loses, and cannot resolve in a controlled sensor-to-brake comparison. Protocol-inspired cases do not constitute compliance with ISO forward-collision standards [24], [25], UN R152 type approval [26], or a Euro NCAP rating [27]. The contribution is a controlled empirical comparison of late brake-permission policies, not a new fusion primitive.

III. SHARED PIPELINE AND PERMISSION POLICIES

A. Sensors, detector, and information boundary

The ego Tesla Model 3 operates in Town04 with synchronous 0.05-s simulation steps. Only radar points and RGB images inform online target perception; ego state supports path prediction and braking. Actor identity, class truth, target velocity, and ground-truth boxes are not queried for brake decisions. Scenario orchestration and offline labels remain outside this information boundary. A weak radar track cannot be repaired using actor truth.

Table I specifies the shared setup. Dataset splits contain 1872/379/264 boxes and 21.2/16.3/18.0% empty frames, with no exact cross-split image duplicates. Session separation prevents adjacent frames from being interleaved across splits, but all data remain Town04 in-domain. The detector metrics reported here are validation metrics from model selection: precision, recall, mAP₅₀, and mAP_{50:95} are 0.9919, 0.9642,

0.9940, and 0.9495. The separate test split is retained in the dataset audit; neither split establishes out-of-domain AEB performance. Calibrated extrinsics project the selected radar target into the image. Confirmation requires the projected point inside a retained car box. Inference cadence and confirmation expiry use simulation timestamps, not host wall time. The identical 0.35-s hold in both camera policies bridges inference intervals but permits bounded stale confirmation.

B. Risk and three alternative gates

Radar points are filtered by range, road height, and predicted constant-curvature path, clustered by position/velocity, and associated across frames. A confirmed, non-stale, collision-relevant track is selected. For range d , relative radar velocity $v_r < 0$, and ego speed v_e ,

$$v_c = -v_r, \quad \text{TTC} = d/v_c, \quad v_t = \max(0, v_e + v_r). \quad (1)$$

The stopping requirement and margin are

$$d_{\text{req}} = \max\left(d_0, v_e t_r + \frac{v_e^2}{2a_e} - \frac{v_t^2}{2a_t} + d_0\right), \quad m = d - d_{\text{req}}, \quad (2)$$

where $t_r = 0.20$ s, $a_e = 8$, $a_t = 6$ m/s², and $d_0 = 1$ m are nominal simulation risk parameters. Later peak decelerations are logged closed-loop outcomes, not constraints imposed by a_e . A shared staged PID supplies continuous brake commands.

Let B_r denote the radar AEB request, C current camera confirmation, H held confirmation, and E_r the emergency condition, including its latch. The permission policies are

$$B_{\text{radar}} = B_r, \quad (3)$$

$$B_{\text{hard}} = B_r \wedge (C \vee H), \quad (4)$$

$$B_{\text{fallback}} = B_r \wedge (C \vee H \vee E_r). \quad (5)$$

A new emergency acceptance requires a confirmed/non-stale track, age and hit streak ≥ 3 , at least six points, track confidence ≥ 0.70 , absolute path offset ≤ 0.65 m, $\text{TTC} \leq 1.10$ s, and margin $m \leq -2.0$ m. TTC and margin are conjunctive, not alternatives. Once accepted, fallback latches until radar leaves BRAKE. Track confidence here is a tracker score, not a calibrated probability of physical existence.

All gates require B_r : a camera box alone cannot create kinematic risk. Radar-only is therefore the unrestricted reference for the shared request; fallback is not radar-only because it can withhold that request without camera confirmation until stricter emergency conditions hold. The pipeline updates ego path, tracks, target, radar risk, scheduled camera inference, permission, and actuation in that order, then logs the subsequent simulator state.

At an identical input state, the equations imply $B_{\text{hard}} \leq B_{\text{fallback}} \leq B_{\text{radar}}$ for Boolean permission. This is a property of the rules, not a guarantee that the closed-loop outcome sets are nested: different brake histories change speed, subsequent sensor observations, and the timing of threshold crossings. The paired comparison therefore uses separately executed conditions, not a counterfactual replay that assumes identical future measurements after different brake actions.

IV. PROTOCOL, LABELS, AND ANALYSIS

A. Frozen execution and constructed suites

Thresholds, suites, and protocol were repository-tagged before hold-out execution at `safe-fallback-eval-v1`. The adverse cases were designed with knowledge of the rule; their outcomes were not used to select or retune its thresholds. This is a *frozen adverse mechanism hold-out*, not a blind sample of a deployment distribution. Named-condition statistics and severity are post-hoc derivations from frozen logs. They must not be mistaken for newly pre-registered outcomes.

The campaign contains 2,461 runs in 639 isolated CARLA sessions. Repetitions of one named condition execute before the next fresh server process. Both camera policies require `CUDAExecutionProvider`; a provider mismatch or inference error is a technical hard stop. Completed algorithmic FAILs remain evidence; only technical incompleteness is eligible for retry.

Core comprises 66 speed-gap conditions spanning CCRm, CCRb, cut-in and related limits, 29 mixed vehicle/negative regressions, eight lateral edge props, and two central box/barrel hazards: 105 conditions in total. Six weak four-point synthetic-fault conditions are evaluated separately, not included in primary core confusion. Twenty local perturbations vary speed by ± 2 km/h, gap by ± 1 m, cut-in timing by ± 0.1 s, or prop offset by ± 0.15 m. Ten detector-disabled conditions remove camera detections; this models complete detection loss rather than a natural-weather distribution.

The 14-condition adverse hold-out contains three new central props, four vehicle appearances, two edge props, and five persistent synthetic-ghost geometries with 6–10 injected points. Core and hold-out use five repeats; development and perturbation use two and three, respectively. Seed 2026 is recorded. No named condition produced mixed PASS/FAIL outcomes across its repeats: reported condition-level system outcomes are all-PASS or all-FAIL. Repeated runs therefore measure consistency, not independent traffic encounters.

B. Brake prediction, system success, and severity

Frozen YAML supplies expected brake/collision labels. A brake prediction means at least one logged tick with `aeb_override=true`; a collision means any tick with `collision_count>0`. System PASS requires brake and collision expectations to match. Positive non-collision cases also require minimum bumper gap ≥ 0.5 m, with additional lane/target assertions where specified. Thus a brake TP may still collide or violate the stopping-gap criterion.

We report TP/FP/TN/FN, precision/recall, system PASS, and collision counts per named condition. Paired matrices match identical condition identifiers. Wilson intervals [29] are descriptive binomial reference intervals computed from named counts, not coverage guarantees or road-population inference for this nonrandom grid. No significance test treats repetitions as independent. Designed class ratios influence precision, and small hold-out denominators limit interpretation.

The labels concern the intended need for braking in each constructed condition, not whether YOLO predicts the *car*

class. A central non-vehicle prop is therefore a positive hazard even when absent from the detector’s label space; a synthetic ghost with no physical hazard is negative despite a stable radar track. These labels allow semantic rejection and physical hazard relevance to be evaluated separately. Named-condition aggregation also retains distinct failures: insufficient clearance, absent braking, and collision are not interchangeable, and identical PASS rates need not imply identical brake timing.

Pre-impact speed is the last logged ego speed before first collision, a 0.05-s proxy rather than exact contact speed. False-brake duration counts override ticks; peak deceleration excludes initialization, sub-1-km/h, and collision ticks. We also report brake-onset speed/gap and whether ego stops below 1 km/h. Medians over repeats summarize execution, not population severity. Without an occupant or following-vehicle model, these are kinematic descriptors, not injury or secondary-crash risk estimates.

V. RESULTS: BENEFITS, COUNTEREXAMPLES, AND SEVERITY

A. Paired operating points

Table II and Fig. 1 show core and hold-out results. Core fallback passes 101/105 conditions, versus 99 for hard gating and 93 for radar-only. Both camera policies suppress eight edge-prop false brakes (four prop types at two lateral offsets). Radar-only and fallback brake for both central box/barrel hazards; hard gating vetoes them, yielding two extra FN/collision conditions. All policies share one FN at a late cut-out corridor boundary and three high-speed collision conditions. Consequently core collision counts are 3/5/3 by policy, or 15/25/15 repeated runs.

Table III exposes the changes hidden by aggregate rates. On core, hard gating gains eight conditions but loses two relative to radar; fallback retains both recoveries. On hold-out, hard gating gains four conditions relative to fallback and loses none in binary system PASS. All three still collide with the three central props. This does not mean equal mitigation, as the bench severity below demonstrates.

All six separate weak four-point fault conditions trigger radar-only false braking and are blocked by both camera policies. Conversely, fallback accepts four of five high-support hold-out ghosts, blocking only the 0.75-m-offset condition. Vehicle-appearance conditions do not distinguish hard gate from fallback because they remain within the camera’s label space. The policy ordering changes with the mechanisms represented in the suite, not simply with the number of repeated executions.

B. Mechanisms at different pipeline stages

Missing track. The hold-out shopping cart produces at most two path candidates and no confirmed cluster. No radar brake request reaches any gate; a late permission override cannot recover this upstream failure. All policies impact without braking at a median proxy speed of 49.96 km/h.

Semantic veto and late permission. The bench forms a radar track, but no usable car confirmation. Radar-only brakes at 12.624 m; fallback qualifies at 4.295 m; hard gating does not brake. Table IV records pre-impact speeds of 32.57, 54.93, and 59.95 km/h, respectively. Fallback recovers a brake TP, not

TABLE II

PRIMARY NAMED-CONDITION RESULTS. BRACKETED RANGES ARE DESCRIPTIVE WILSON INTERVALS ON THE CONSTRUCTED GRID ONLY; COLL.: COLLISION CONDITIONS.

Scope	Policy	N	TP	FP	TN	FN	Precision [range]	Recall [range]	PASS	Coll.
Core	Radar	105	84	8	12	1	.913 [.838,.955]	.988 [.936,.998]	93	3
	Hard	105	82	0	20	3	1.000 [.955,1]	.965 [.901,.988]	99	5
	Fallback	105	84	0	20	1	1.000 [.956,1]	.988 [.936,.998]	101	3
Hold-out	Radar	14	6	5	2	1	.545 [.280,.787]	.857 [.487,.974]	6	3
	Hard	14	5	0	7	2	1.000 [.566,1]	.714 [.359,.918]	11	3
	Fallback	14	6	4	3	1	.600 [.313,.832]	.857 [.487,.974]	7	3

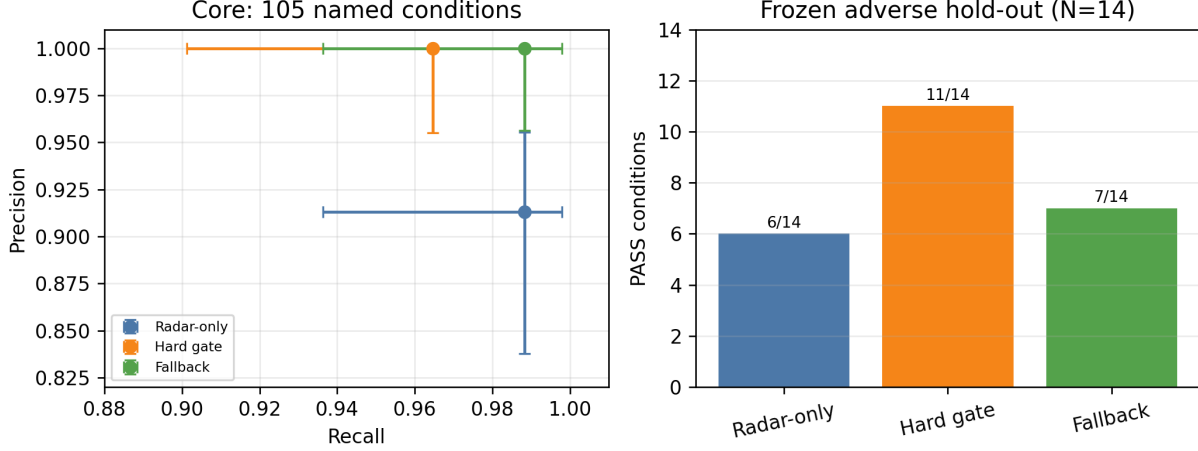


Fig. 1. Frozen named-condition evidence. Left: core precision-recall with descriptive Wilson ranges on constructed named conditions. Right: system PASS on the 14-condition adverse hold-out. Neither grid estimates road prevalence.

TABLE III

PAIRED SYSTEM OUTCOMES BY NAMED CONDITION. COLUMNS REFER TO POLICIES A AND B, NOT BRAKE TP/FP.

Scope	A / B	Both PASS	A only	B only	Both FAIL
Core	Radar / Hard	91	2	8	4
	Radar / Fallback	93	0	8	4
	Hard / Fallback	99	0	2	4
Hold-out	Radar / Hard	6	0	5	3
	Radar / Fallback	6	0	1	7
	Hard / Fallback	7	4	0	3

TABLE IV

HOLD-OUT SEVERITY: MEDIAN OVER FIVE RUNS PER PHYSICAL CONDITION; GHOST ROWS AGGREGATE 25/20 RUNS. PRE-IMPACT IS A LAST-TICK PROXY.

Mechanism	Policy	Onset / gap	Severity outcome
Cart Bench	All	no brake	49.96 km/h pre-impact
	Radar	12.624 m	32.57 km/h pre-impact
	Hard	no brake	59.95 km/h pre-impact
	Fallback	4.295 m	54.93 km/h pre-impact
Warning	All	21.871 m	7.82 km/h pre-impact
Ghost FP	Radar (25 runs)	78.4 km/h	3.60 s / 9.02 m/s ²
	Fallback (20)	78.4 km/h	3.60 s / 9.02 m/s ²

timely stopping. Relative to hard gating, radar-only lowers this proxy by about 27.4 km/h, compared with only 5.0 km/h for fallback.

Collision despite permission. For the traffic-warning prop, all policies brake at 21.871 m, and camera-policy logs show confirmation rather than fallback. All collide at 7.82 km/h. The record establishes that permission was granted; it does not independently identify a contact-dynamics or controller defect. It demonstrates mitigation hidden by a binary collision count.

Convincing false track. A six-point center injection forms a confirmed track by 0.15 s with confidence 1.0, TTC 0.89 s, and margin -15.21 m. Radar-only and fallback request emergency braking; hard gating has no car confirmation. Eight- and ten-point center/0.50-m ghosts behave similarly. Moving the injection to 0.75 m crosses the frozen 0.65-m fallback boundary but does not prevent radar-only braking. These are

synthetic counterexamples to the sufficiency of the emergency rule, not measurements of real ghost prevalence.

Every ghost false brake is a simulated stop below 1 km/h, not a one-tick pulse. The ghost rows report onset speed, override duration, and peak deceleration; these observations make the precision loss operationally meaningful within the simulation. They do not quantify injury or rear-end collision risk. The shared late cut-out FN and high-speed stopping-limit collisions are different again: they are not evidence of a camera veto, because radar-only also fails them.

C. Ablation, perturbations, and detection loss

In the 12-condition development ablation (two repeats), full fallback passes 12/12. Hard gate and no-latch each fail two central box/barrel conditions. Removing the path constraint

TABLE V
EXTENDED OUTCOMES BY NAMED CONDITION, NOT REPEATED RUN.

Suite	Policy	Pass/N	TP	FP	TN	FN	Coll.
Perturb.	Radar	19/20	20	0	0	0	1
	Hard	15/20	15	0	0	5	5
	Fallback	18/20	20	0	0	0	1
Camera off	Hard	4/10	0	0	4	6	6
	Fallback	8/10	6	0	4	0	2

adds three edge-prop false brakes; removing minimum point count adds two weak-ghost false brakes. Removing stability/confidence or replacing TTC-AND-margin by OR changes no outcome on this focused grid. Their independent necessity is therefore not established. Six one-factor sensitivity conditions show a false brake when point count is lowered and a missed central prop when stability or TTC is tightened. No global optimum is claimed.

Table V separates two perturbation failures: fallback collides at the 19-m box gap, but avoids collision at 21 m and stops at 0.353 m, below the 0.5-m PASS threshold. Equating every FAIL with collision would be incorrect. With detections disabled, fallback brakes for all six positive conditions but collides in two. Hard gate’s four PASS conditions are all negatives, not successful hazard responses. These local probes expose rule dependence, not robustness across maps, weather, or stochastic sensor errors.

D. Computation and reproducibility

The 474 required-CUDA sessions contain 74,928 inferences and zero inference errors. Session-median p50/p95 are 9.95/11.53 ms; the inference-weighted mean is 11.56 ms. Each fresh process has one cold start above 150 ms; the maximum is 276.10 ms. These host-measured ONNX processing times exclude radar, transport, control, display, and actuation; they are not end-to-end ECU deadlines.

Session manifests record commit, seed, CARLA versions, model/config hashes, configured/active providers, inference counts, and errors. Tick logs retain track support, confidence, TTC, margin, gate reason, override, and collision. Checkpoint keys are (scenario identifier, run index); technical recovery cannot turn an algorithmic FAIL into a selected successful retry. Named confusion, paired outcomes, and severity regenerate from the frozen logs without adding or removing a run. Repository refactoring and launcher changes occurred later and are not experimental contributions or new validation of these policies.

Source code, manuscript files, checksums, and curated derived evidence are indexed at <https://github.com/mvhoang92/aeb>; the companion source map identifies the frozen raw archive, derivation scripts, and checksum trail. Large runtime artifacts and local model files are referenced by hash and path policy rather than embedded in the manuscript package. Re-execution requires a compatible CARLA installation, local model artifact, and CUDA environment. Hashes establish artifact identity, not hardware-independent bitwise behavior. The provenance

distinction matters: regression testing of later software does not enlarge the scientific sample or replace the archived campaign.

VI. DISCUSSION AND VALIDITY

A. What the comparison establishes

The results locate three different questions: did radar form a target, did the policy permit braking, and did the permitted action achieve sufficient mitigation? Collapsing them into one success rate obscures both the bench’s late permission and the warning prop’s low-speed collision. Likewise, the emergency rule improves core recall but admits ghosts with the same qualifying attributes. Track persistence and point support strengthen an estimate without independently proving a physical obstacle exists.

This is a limitation of the tested information and thresholds, not an impossibility theorem for radar-camera fusion. Hard gating is favored by binary hold-out PASS here, but not by bench mitigation; fallback is favored on core, but its hold-out false brakes are full stops. We do not prescribe a universal policy by assigning arbitrary weights to these incompatible outcomes. A useful next comparison must include class-agnostic visual verification and a confidence-matched soft gate before attributing gains to a new fusion algorithm. Probabilistic methods require calibration and evidence beyond a renamed tracker score [22], [23].

The resulting answers to RQ1–RQ3 are conditional but actionable. Policy gains can be traced to a small number of identified mechanisms rather than a generic “fusion improvement.” An override cannot repair the absence of an upstream track, and passing a radar-quality threshold can both admit a ghost and delay a real obstacle. Finally, collision avoidance, impact mitigation, and unwanted intervention are distinct objectives. A policy that wins on one should not be selected from a single aggregate F1 or PASS rate without specifying the intended operating conditions and acceptable error costs. The present suite supports these comparisons within its grid, not deployment-specific cost estimation.

B. Limits of the evidence

The experiment covers one map, ego platform, fixed seed, favorable imaging, and an in-domain car-only detector. Repeats are correlated, and the 14-condition hold-out was deliberately constructed with rule knowledge. It tests adverse mechanisms rather than unbiased generalization. Point-level CARLA radar does not reproduce full multipath physics; fault injection does not estimate real radar-ghost frequency [19], [21]. The last pre-impact tick has 0.05-s discretization, and false-brake kinematics are not injury costs. There is no multi-map/weather/friction sweep, real-sensor experiment, HIL, functional-safety demonstration, or standards certification.

There is also no matched soft-gate, class-agnostic, learned, or probabilistic baseline. The camera-veto limitation is conditional on the chosen car-only label space, not a claim about all camera systems. Expanded baselines require a new development/evaluation protocol; the already inspected hold-out may remain regression data but cannot remain unseen test evidence.

This restriction preserves the present adverse results instead of tuning away the counterexamples.

VII. CONCLUSION

In a fixed CARLA pipeline, emergency fallback recovers two camera-vetoed core hazards while retaining edge-prop false-brake suppression. Four persistent synthetic-ghost conditions reverse its advantage over hard gating on adverse hold-out. Full-stop false brakes and delayed bench mitigation show why brake recall and collision counts alone are insufficient. The evidence supports a mechanism- and severity-aware comparison of permission policies, not a general fusion advantage. Missing camera confirmation is not proof of free road, and stable radar support is not proof of a physical hazard.

REFERENCES

- [1] J. B. Cicchino, "Effectiveness of forward collision warning and autonomous emergency braking systems in reducing front-to-rear crash rates," *Accident Analysis & Prevention*, vol. 99, pp. 142–152, 2017.
- [2] E. Coelingh, A. Eidehall, and M. Bengtsson, "Collision warning with full auto brake and pedestrian detection—a practical example of automatic emergency braking," in *13th International IEEE Conference on Intelligent Transportation Systems (ITSC)*, 2010, pp. 155–160.
- [3] L. Yang, Y. Yang, G. Wu, X. Zhao, S. Fang, X. Liao, R. Wang, and M. Zhang, "A systematic review of autonomous emergency braking system: Impact factor, technology, and performance evaluation," *Journal of Advanced Transportation*, vol. 2022, pp. 1–13, 2022.
- [4] International Organization for Standardization, "Road vehicles—safety of the intended functionality," ISO 21448:2022, 2022. [Online]. Available: <https://www.iso.org/standard/77490.html>
- [5] A. Dosovitskiy, G. Ros, F. Codevilla, A. López, and V. Koltun, "CARLA: An open urban driving simulator," in *Proceedings of the 1st Conference on Robot Learning*, 2017, pp. 1–16.
- [6] Y. Kang, H. Yin, and C. Berger, "Test your self-driving algorithm: An overview of publicly available driving datasets and virtual testing environments," *IEEE Transactions on Intelligent Vehicles*, vol. 4, no. 2, pp. 171–185, 2019.
- [7] H. Kim and B. Song, "Vehicle recognition based on radar and vision sensor fusion for automatic emergency braking," in *2013 13th International Conference on Control, Automation and Systems*, 2013, pp. 1342–1346.
- [8] P.-M. Hsu, M. H. Li, and K.-C. Chang, "Noise filtering in autonomous emergency braking systems with sensor fusions," SAE International, Tech. Rep. 2015-01-0216, 2015.
- [9] H. Bae, G. Lee, J. Yang, G. Shin, G. Choi, and Y. Lim, "Estimation of the closest in-path vehicle by low-channel LiDAR and camera sensor fusion for autonomous vehicles," *Sensors*, vol. 21, no. 9, p. 3124, 2021.
- [10] L. Zhang, Z. Yu, X. Xu, and Y. Yan, "Research on automatic emergency braking system based on target recognition and fusion control strategy in curved road," *Electronics*, vol. 12, no. 16, p. 3490, 2023.
- [11] F. Nobis, M. Geisslinger, M. Weber, J. Betz, and M. Lienkamp, "A deep learning-based radar and camera sensor fusion architecture for object detection," in *2019 Sensor Data Fusion: Trends, Solutions, Applications (SDF)*, 2019, pp. 1–7.
- [12] S. Chadwick, W. Maddern, and P. Newman, "Distant vehicle detection using radar and vision," in *2019 International Conference on Robotics and Automation (ICRA)*, 2019, pp. 8311–8317.
- [13] R. Nabati and H. Qi, "CenterFusion: Center-based radar and camera fusion for 3D object detection," in *2021 IEEE Winter Conference on Applications of Computer Vision*, 2021, pp. 1526–1535.
- [14] R. C. Akula, S. Kandhasamy, and M. Ganesan, "Radar-guided camera verification for automatic emergency braking: Rethinking object detection in radar-camera fusion," *arXiv preprint arXiv:2606.27556*, 2026. [Online]. Available: <https://arxiv.org/abs/2606.27556>
- [15] S. Riedmaier, T. Ponn, D. Ludwig, B. Schick, and F. Diermeyer, "Survey on scenario-based safety assessment of automated vehicles," *IEEE Access*, vol. 8, pp. 87 456–87 477, 2020.
- [16] C. Gómez-Huélamo, J. D. Egidio, L. M. Bergasa, R. Barea, E. López-Guillén, F. Arango, J. Araluce, and J. López, "Train here, drive there: ROS-based end-to-end autonomous-driving pipeline validation in CARLA simulator using the NHTSA typology," *Multimedia Tools and Applications*, vol. 81, no. 3, pp. 4213–4240, 2022.
- [17] J. Feng, P. Zhao, J. Konpang, A. Sirikham, H. Zheng, P. Kalnaowakul, and J. Wang, "Validating DVS application in autonomous driving with various AEB scenarios in CARLA simulator," *World Electric Vehicle Journal*, vol. 16, no. 11, p. 634, 2025.
- [18] R. Rao, C. Cui, L. Chen, T. Gao, and Y. Shi, "Quantitative testing and analysis of non-standard AEB scenarios extracted from corner cases," *Applied Sciences*, vol. 14, no. 1, p. 173, 2024, published online December 24, 2023.
- [19] F. Kraus, N. Scheiner, W. Ritter, and K. Dietmayer, "The radar ghost dataset—an evaluation of ghost objects in automotive radar data," in *2021 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2021, pp. 8570–8577.
- [20] S. Jha, S. Banerjee, T. Tsai, S. K. S. Hari, M. B. Sullivan, Z. T. Kalbarczyk, S. W. Keckler, and R. K. Iyer, "ML-based fault injection for autonomous vehicles: A case for Bayesian fault injection," in *2019 49th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN)*, 2019, pp. 112–124.
- [21] Z. F. Magosi, H. Li, P. Rosenberger, L. Wan, and A. Eichberger, "A survey on modelling of automotive radar sensors for virtual test and validation of automated driving," *Sensors*, vol. 22, no. 15, p. 5693, 2022.
- [22] F. Sandblom and M. Brännström, "Probabilistic threat assessment and driver modeling in collision avoidance systems," in *2011 IEEE Intelligent Vehicles Symposium (IV)*, 2011, pp. 914–919.
- [23] L. Lindenmaier, S. Aradi, T. Bécsi, O. Törő, and P. Gáspár, "GM-PHD filter based sensor data fusion for automotive frontal perception system," *IEEE Transactions on Vehicular Technology*, vol. 71, no. 7, pp. 7215–7229, 2022.
- [24] *Transport Information and Control Systems—Forward Vehicle Collision Warning Systems—Performance Requirements and Test Procedures*, International Organization for Standardization Std. ISO 15 623:2013, 2013.
- [25] International Organization for Standardization, "Intelligent transport systems—forward vehicle collision mitigation systems—operation, performance, and verification requirements," ISO 22839:2013, 2013. [Online]. Available: <https://www.iso.org/standard/45339.html>
- [26] United Nations Economic Commission for Europe, "UN Regulation No. 152: Uniform provisions concerning the approval of motor vehicles with regard to the advanced emergency braking system (AEBS) for M1 and N1 vehicles," Official Journal of the European Union, L 360, pp. 66–89, 2020, entered into force 22 January 2020. [Online]. Available: <https://eur-lex.europa.eu/eli/reg/2020/1597/oj/eng>
- [27] Euro NCAP, "Test protocol—AEB car-to-car systems," Version 4.2, Implementation 2023, June 2023, 2023. [Online]. Available: https://cdn.euroncap.com/cars/assets/euro_ncap_aeb_c2c_test_protocol_v42_af2bd7186c.pdf
- [28] Ultralytics, "Ultralytics YOLO documentation," accessed: 2026-08-18. [Online]. Available: <https://docs.ultralytics.com/>
- [29] E. B. Wilson, "Probable inference, the law of succession, and statistical inference," *Journal of the American Statistical Association*, vol. 22, no. 158, pp. 209–212, 1927.